

Homework 3 | Solutions

Name(s): [enter name(s)]

Excel uploader: [enter name]

Problem 1(a)-(c) | Construction cost only

Total demand is $1,000 + 3,000 + 3,000 + 1,000 + 4,000 = 12,000$ cars. The facility data used in all three models are:

Plant	Fixed cost (\$)	Cost/car (\$)	Minimum	Capacity
1	2,100,000	380	2,000	5,000
2	2,800,000	165	3,000	7,000
3	2,400,000	155	4,000	8,000
4	2,700,000	170	2,500	6,000

(a) Integer programming formulation

Let $y_i = 1$ if Plant i is built, and 0 otherwise, for $i = 1, 2, 3, 4$. Let F_i and U_i denote the fixed cost and maximum capacity in the table.

Minimize $Z = \sum_i F_i y_i$

$= 2,100,000y_1 + 2,800,000y_2 + 2,400,000y_3 + 2,700,000y_4$

Subject to: $5,000y_1 + 7,000y_2 + 8,000y_3 + 6,000y_4 \geq 12,000$

$y_i \in \{0,1\}$ for every plant i .

This first model selects sufficient installed capacity. It does not choose production quantities, so minimum production and variable costs do not enter this model.

(b)-(c) Solver result and recommendation

Build Plants 1 and 3. The minimum construction cost is \$4,500,000. Their combined capacity is 13,000 cars, leaving capacity for 1,000 cars beyond forecast demand. Do not build Plants 2 and 4.

No single plant can meet demand. Plants 1 and 3 form the least costly feasible pair, and every three-plant combination costs more.

Problem 1: Min B27; change B22:B25 (binary); B29 $\geq$ 0.

C22=B22*E5; D22=B22*B5, copied down to row 25.

B27=SUM(D22:D25); B28=SUM(C22:C25); B29=B28-D28; D28=B17.

Problem 1(b) | Excel evidence

Actual Excel worksheet, Solver Parameters, and Solver Results screenshots. The formulas and settings are documented in the workbook and on the preceding solution page.

	A	B	C	D
20	(a)-(c) Construction cost only			
21	Facility	Build (binary)	Capacity	Fixed cost (\$)
22	Plant 1	1	5000	\$2,100,000
23	Plant 2	0	0	\$0
24	Plant 3	1	8000	\$2,400,000
25	Plant 4	0	0	\$0
26				
27	Construction cost	\$4,500,000		
28	Total capacity	13000	>=	12000
29	Capacity slack	1000		

Solver Parameters

Set Objective:

To: ☐ Max ☒ Min ☐ Value Of:

By Changing Variable Cells:

Subject to the Constraints:

☒ Make Unconstrained Variables Non-Negative

Select a Solving Method:

Solving Method
Select the GRG Nonlinear engine for Solver Problems that are smooth nonlinear. Select the LP Simplex engine for linear Solver Problems, and select the Evolutionary engine for Solver problems that are non-smooth.

Solver Results

Solver found an integer solution within tolerance. All constraints are satisfied.

☒ Keep Solver Solution
☐ Restore Original Values

Method: Simplex LP. Integer optimality: 0%. Integer constraints enforced. Unconstrained variables nonnegative.

Problem 1(d)-(f) | Construction and production

(d) Integer programming formulation

Let y_i be the binary build decision defined in part (a), and let q_i be the nonnegative integer number of cars produced at Plant i . Let c_i and L_i denote production cost per car and minimum production if built.

$$\text{Minimize } Z = \sum_i F_i y_i + \sum_i c_i q_i$$

$$\text{Subject to: } \sum_i q_i \geq 12,000$$

$$L_i y_i \leq q_i \leq U_i y_i \quad \text{for } i = 1, 2, 3, 4$$

$$y_i \in \{0,1\}; q_i \in \{0,1,2,\dots\}.$$

The linking constraints force production to zero at an unbuilt plant and enforce both production limits at a built plant. All production costs are positive, and this optimum produces exactly 12,000 cars.

(e)-(f) Solver result and recommendation

Plant	Build?	Cars produced	Production cost
1	No	0	\$0
2	No	0	\$0
3	Yes	8,000	\$1,240,000
4	Yes	4,000	\$680,000

Build Plants 3 and 4. Produce 8,000 cars at Plant 3 and 4,000 at Plant 4. Construction costs are \$5,100,000 and production costs are \$1,920,000, for a minimum total cost of \$7,020,000.

Plant 3 meets its 4,000-car minimum and reaches its 8,000-car capacity. Plant 4 produces between its 2,500-car minimum and 6,000-car capacity. Total unused capacity is 2,000 cars.

Excel implementation

Min B43; change B38:C41. B38:B41 binary; C38:C41 integer.
Constraints: H38:I41 >= 0; B47 >= 0; nonnegative variables.

D38=B38*D5; E38=B38*E5; F38=B38*B5; G38=C38*C5.
H38=C38-D38; I38=E38-C38. Copy through row 41.
B43=SUM(F38:G41); B46=SUM(C38:C41); B47=B46-D46.
D46=B17, where B17=SUM(B16:F16)=12000.

H and I are the lower-bound and capacity slacks. Requiring both to be nonnegative implements the linking constraints exactly.

Problem 1(e) | Excel evidence

Actual Excel worksheet, Solver Parameters, and Solver Results screenshots. The formulas and settings are documented in the workbook and on the preceding solution page.

	A	B	C	D	E	F	G	H	I
36	(d)-(f) Construction plus production costs								
37	Facility	Build (binary)	Cars (integer)	Min if built	Max if built	Fixed cost (\$)	Production (\$)	Min slack	Capacity slack
38	Plant 1	0	0	0	0	\$0	\$0	0	0
39	Plant 2	0	0	0	0	\$0	\$0	0	0
40	Plant 3	1	8000	4000	8000	\$2,400,000	\$1,240,000	4,000	0
41	Plant 4	1	4000	2500	6000	\$2,700,000	\$680,000	1,500	2,000
42									
43	Total cost	\$7,020,000							
44	Construction cost	\$5,100,000							
45	Production cost	\$1,920,000							
46	Total cars	12000 >=		12000					
47	Demand slack	0							

Solver Parameters

Set Objective:

To: ☐ Max ☒ Min Value Of:

By Changing Variable Cells:

Subject to the Constraints:

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☒ Make Unconstrained Variables Non-Negative

Select a Solving Method:

Solving Method
Select the GRG Nonlinear engine for Solver Problems that are smooth nonlinear. Select the LP Simplex engine for linear Solver Problems, and select the Evolutionary engine for Solver problems that are non-smooth.

Solver Results

Solver found a solution. All constraints and optimality conditions are satisfied.

☒ Keep Solver Solution ☐ Restore Original Values

☐ Return to Solver Parameters Dialog ☐ Outline Reports

Method: Simplex LP. Integer optimality: 0%. Integer constraints enforced. Unconstrained variables nonnegative.

Problem 1(g) | Full cost model

Let x_{ij} be the nonnegative integer number of cars shipped from Plant i to city j . Retain the build decisions y_i and production quantities q_i . Each produced car is shipped to one of the five cities.

\$/car	Pittsburgh	Chicago	New York	D.C.	Boston
Plant 1	18	35	50	30	20
Plant 2	15	10	12	10	12
Plant 3	40	50	30	35	25
Plant 4	32	25	20	25	40
Demand	1,000	3,000	3,000	1,000	4,000

Let t_{ij} be the transportation cost in the table and d_j the demand of city j . The city order is Pittsburgh, Chicago, New York, Washington, D.C., Boston.

$$\text{Minimize } Z = \sum_i F_i y_i + \sum_i c_i q_i + \sum_i \sum_j t_{ij} x_{ij}$$

Subject to: $\sum_i x_{ij} = d_j$ for every city j

$$q_i = \sum_j x_{ij} \text{ for every plant } i$$

$$L_i y_i \leq q_i \leq U_i y_i \text{ for every plant } i$$

$y_i \in \{0,1\}$; q_i and x_{ij} are nonnegative integers.

City constraints meet each demand exactly. The production balances ensure that a plant produces what it ships. In Excel, q_i is calculated as the row sum of shipments, so it does not need a separate changing cell.

Excel implementation for part (h)

Min B66; change B57:G60. B57:B60 binary; C57:G60 integer.

Constraints: K57:L60 ≥ 0 ; C64:G64 = 0; nonnegative variables.

H57=SUM(C57:G57); I57=B57*D5; J57=B57*E5.

K57=H57-I57; L57=J57-H57. Copy through row 60.

C62=SUM(C57:C60); C63=B16; C64=C62-C63.

Copy the city formulas across through column G.

B67=SUMPRODUCT(B57:B60,B5:B8)

B68=SUMPRODUCT(H57:H60,C5:C8)

B69=SUMPRODUCT(C57:G60,B12:F15)

B66=SUM(B67:B69)

Problem 1(h)-(i) | Full cost solution

Build Plants 2 and 3. Produce 7,000 cars at Plant 2 and 5,000 cars at Plant 3. The minimum total cost is \$7,339,000.

From	Pittsburgh	Chicago	New York	D.C.	Boston
Plant 1	0	0	0	0	0
Plant 2	1,000	3,000	2,000	1,000	0
Plant 3	0	0	1,000	0	4,000
Plant 4	0	0	0	0	0
Total	1,000	3,000	3,000	1,000	4,000

Cost calculation

Construction = $2,800,000 + 2,400,000 = \$5,200,000$

Production = $7,000(165) + 5,000(155) = \$1,930,000$

Plant 2 transport = $1,000(15) + 3,000(10) + 2,000(12) + 1,000(10)$
 $= \$79,000$

Plant 3 transport = $1,000(30) + 4,000(25) = \$130,000$

Total = $5,200,000 + 1,930,000 + 209,000 = \$7,339,000$

All city demands are met. Plant 2 uses its full 7,000-car capacity. Plant 3 produces above its 4,000-car minimum and has 3,000 cars of unused capacity. Plants 1 and 4 are not built and have zero production and shipments.

Recommendation to the company

Use the full cost model for the final facility decision. Plants 2 and 3 cost \$100,000 more to build and \$10,000 more to operate than Plants 3 and 4 at their production-only optimum, but Plant 2 has much lower shipping costs. Optimizing all costs together selects Plants 2 and 3.

The three reported objective values cover different cost categories. They should not be compared as if they were three estimates of the same total cost.

Problem 1(h) | Excel evidence

Actual Excel worksheet, Solver Parameters, and Solver Results screenshots. The formulas and settings are documented in the workbook and on the preceding solution page.

(g)-(i) Construction, production and transportation costs											
Facility	Build (binary)	Pittsburgh	Chicago	New York	Washington, D.C.	Boston	Production	Min if built	Max if built	Min slack	Capacity slack
Plant 1	0	0	0	0	0	0	0	0	0	0	0
Plant 2	1	1,000	3,000	2,000	1,000	0	0	7,000	3,000	7,000	4,000
Plant 3	1	0	0	1,000	0	0	4,000	5,000	4,000	8,000	1,000
Plant 4	0	0	0	0	0	0	0	0	0	0	0
Delivered		1000	3000	3000	1000	4000					
Demand		1000	3000	3000	1000	4000					
Demand residual		0	0	0	0	0					
Total cost		\$7,339,000									
Construction cost		\$5,200,000									
Production cost		\$1,930,000									
Transportation cost		\$209,000									

Solver Parameters

Set Objective:

To: ☐ Max ☒ Min ☐ Value Of:

By Changing Variable Cells:

Subject to the Constraints:

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☒ Make Unconstrained Variables Non-Negative

Select a Solving Method:

Solving Method
Select the GRG Nonlinear engine for Solver Problems that are smooth nonlinear. Select the LP Simplex engine for linear Solver Problems, and select the Evolutionary engine for Solver problems that are non-smooth.

Solver Results

Solver found a solution. All constraints and optimality conditions are satisfied.

☒ Keep Solver Solution ☐ Restore Original Values

☐ Return to Solver Parameters Dialog ☐ Outline Reports

Reports

Method: Simplex LP. Integer optimality: 0%. Integer constraints enforced. Unconstrained variables nonnegative.

Problem 2(a) | Match scheduling model

Use the assignment's hypothetical group and visibility data. Let $M = \{1, \dots, 6\}$ be the matches and $D = \{1, \dots, 14\}$ be the available days. Match numbers identify pairings; they do not require chronological order.

Match	Pairing
1	Colombia vs. Germany (C-G)
2	Colombia vs. Philippines (C-P)
3	New Zealand vs. Germany (NZ-G)
4	New Zealand vs. Philippines (NZ-P)
5	Colombia vs. New Zealand (C-NZ)
6	Germany vs. Philippines (G-P)

Define $x_{md} = 1$ if match m is scheduled on day d , and 0 otherwise. Let v_{md} be its visibility score. For each team t , M_t is the set of matches involving that team:

$$M_C = \{1, 2, 5\}; M_{NZ} = \{3, 4, 5\}; M_G = \{1, 3, 6\}; M_P = \{2, 4, 6\}.$$

$$\text{Maximize } V = \sum_m \sum_d v_{md} x_{md}$$

$$\sum_d x_{md} = 1 \quad \text{for every match } m \quad [\text{play each match exactly once}]$$

$$\sum_m x_{md} \leq 2 \quad \text{for every day } d \quad [\text{at most two matches per day}]$$

$$\sum_{(m \in M_t)} x_{md} \leq 1 \quad \text{for every team } t \text{ and day } d$$

$$\sum_m x_{m,14} = 2 \quad [\text{exactly two matches on the last day}]$$

$$x_{md} \in \{0, 1\} \quad \text{for all matches and days.}$$

The team constraints make the two day-14 matches disjoint, so every team plays once on the final day. Give those two matches the same kickoff time. Since this model selects days only, their common kickoff time is an operational requirement rather than a separate day-assignment variable.

"All matches should be played on the 14 days" is interpreted as scheduling all six matches within days 1 through 14. It cannot mean that each of the 14 days must contain a match.

Problem 2(b)-(c) | Optimal schedule

Day	Match	Visibility
5	New Zealand vs. Philippines	177
9	Colombia vs. Germany	192
13	Colombia vs. New Zealand	188
13	Germany vs. Philippines	181
14	Colombia vs. Philippines	150
14	New Zealand vs. Germany	150

Maximum visibility = $177 + 192 + 188 + 181 + 150 + 150 = 1,038$.

Recommend the schedule above. Both day-14 games must begin at the same time. No matches are scheduled on the other days. Every match appears once, each team plays three matches, no team plays twice on one day, and no day has more than two games.

Colombia and Germany play on days 9, 13 and 14. New Zealand and the Philippines play on days 5, 13 and 14. Thus, all four teams play on consecutive days 13 and 14. This is allowed in parts (a)-(c), but motivates the added constraint in part (d).

Excel implementation

Problem 2: Max B43; change B24:G37 (binary).
 B39:G39 = 1; H24:H37 <= 2; I24:L37 <= 1; H37 = 2.

H24=SUM(B24:G24)
 I24=B24+C24+F24; J24=D24+E24+F24
 K24=B24+D24+G24; L24=C24+E24+G24
 Copy these daily formulas through row 37.

B39=SUM(B24:B37)
 B40=SUMPRODUCT(\$A\$24:\$A\$37,B24:B37)
 B41=SUMPRODUCT(B6:B19,B24:B37)
 Copy these match formulas across through column G.
 B43=SUMPRODUCT(B6:G19,B24:G37)

The visibility input matrix occupies B6:G19. The binary decision matrix uses the same day and match order in B24:G37. The workbook also includes a three-day-window check for part (d).

Problem 2(b) | Excel evidence

Actual Excel worksheet, Solver Parameters, and Solver Results screenshots. The formulas and settings are documented in the workbook and on the preceding solution page.

	A	B	C	D	E	F	G	H	I	J	K	L
22	(a)-(c) Optimal schedule: 1 means the match is played on that day											
23	Day	C-G	C-P	NZ-G	NZ-P	C-NZ	G-P	Games	Colombia	New Zealand	Germany	Philippines
24	1	0	0	0	0	0	0	0	0	0	0	0
25	2	0	0	0	0	0	0	0	0	0	0	0
26	3	0	0	0	0	0	0	0	0	0	0	0
27	4	0	0	0	0	0	0	0	0	0	0	0
28	5	0	0	0	1	0	0	1	0	1	0	1
29	6	0	0	0	0	0	0	0	0	0	0	0
30	7	0	0	0	0	0	0	0	0	0	0	0
31	8	0	0	0	0	0	0	0	0	0	0	0
32	9	1	0	0	0	0	0	1	1	0	1	0
33	10	0	0	0	0	0	0	0	0	0	0	0
34	11	0	0	0	0	0	0	0	0	0	0	0
35	12	0	0	0	0	0	0	0	0	0	0	0
36	13	0	0	0	0	1	1	2	1	1	1	1
37	14	0	1	1	0	0	0	2	1	1	1	1
38												
39	Match count	1	1	1	1	1	1					
40	Match day	9	14	14	5	13	13					
41	Visibility	192	150	150	177	188	181					
42												
43	Total visibility	1,038										

Solver Parameters

Set Objective:

To: ☒ Max ☐ Min ☐ Value Of:

By Changing Variable Cells:

Subject to the Constraints:

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☒ Make Unconstrained Variables Non-Negative

Select a Solving Method:

Solving Method
Select the GRG Nonlinear engine for Solver Problems that are smooth nonlinear. Select the LP Simplex engine for linear Solver Problems, and select the Evolutionary engine for Solver problems that are non-smooth.

Solver Results

Solver found a solution. All constraints and optimality conditions are satisfied.

☒ Keep Solver Solution ☐ Restore Original Values

☐ Return to Solver Parameters Dialog ☐ Outline Reports

Method: Simplex LP. Integer optimality: 0%. Integer constraints enforced. Unconstrained variables nonnegative.

Problem 2(d) | At least two days of rest

Add the following rolling three-day-window constraint for every team t and every starting day $k = 1, \dots, 12$:

$$\sum_{m \in M_t} [x_{m,k} + x_{m,k+1} + x_{m,k+2}] \leq 1.$$

A team may play at most one match in any three consecutive days. Therefore, a match on day 1 excludes days 2 and 3, and the next match can occur on day 4 or later. This is a linear constraint in the original binary variables and requires no new decision variables.

Examples

For Colombia, $k = 1$:

$$(x_{1,1} + x_{2,1} + x_{5,1}) + (x_{1,2} + x_{2,2} + x_{5,2}) + (x_{1,3} + x_{2,3} + x_{5,3}) \leq 1.$$

For $k = 12$, the constraint covers days 12, 13 and 14. Since every team plays on day 14, no team can play on day 12 or 13 under the new rule. The original optimal schedule is therefore infeasible once this rest requirement is added.

Excel implementation

B61=SUM(I24:I26), copied across through E61 and down through row 72.
Add B61:E72 <= 1 to the existing Problem 2 Solver model.

Columns I:L contain each team's daily match count. The 12 rows of the rest check represent windows 1-3, 2-4, ..., 12-14. Part (d) asks for the mathematical constraint, so the workbook retains the requested base optimum for parts (b)-(c).

Problem 2 | Visibility data and source notes

Day	C-G	C-P	NZ-G	NZ-P	C-NZ	G-P
1	185	170	183	168	178	175
2	130	125	125	120	125	125
3	110	120	115	125	115	120
4	138	140	138	140	140	140
5	175	170	182	177	177	175
6	120	125	125	130	125	125
7	110	110	110	110	110	110
8	144	142	142	140	142	142
9	192	183	177	168	175	185
10	130	130	130	130	130	130
11	120	120	120	120	120	120
12	156	153	153	150	153	153
13	180	175	182	175	188	181
14	150	150	150	150	150	150

Source: HW3.pdf, Problem 1 on pages 1-2 and Problem 2 on pages 2-3. All homework input values were transcribed from this file. All screenshots show the actual Excel workbook or Excel Solver dialogs. Screenshot crops remove unrelated desktop content only.

Workbook organization and reproducibility

HW3_Solved.xlsx contains one visible tab per problem. Problem 1 contains all three cost models. Each model has its objective, decision cells, constraints, and formulas documented next to it.

Saved Solver models are stored in hidden column M on Problem 1 and hidden column N on Problem 2. Use Solver's Load/Save command to load M20:M25 for construction only, M48:M55 for production, M82:M89 for transportation, or N5:N13 on Problem 2 for scheduling. Unhide the column if you want to inspect the saved model cells.

All models use Simplex LP with binary and integer restrictions, nonnegative variables, automatic scaling, and integer optimality tolerance of 0%. The "Ignore Integer Constraints" option is off. The reported objective values were independently checked using a separate integer optimization implementation.